

Henry Stevens

Boulder, CO 80302 henry.stevens321@gmail.com (719)822-5705

EDUCATION

UNIVERSITY OF COLORADO BOULDER

August 2022 - May 2026

Mechanical Engineering Bachelor of Science — Overall GPA: 3.993/4.00

Minor: Computer Science

Certifications: Fundamentals of Engineering (FE) Certification, CSWA-Mechanical Design Certification

- Relevant Coursework: Design for Manufacturability, Mechatronics and Robotics I and II, Control Systems Analysis, Component Design, CAD and Fabrication, Computational Methods, Computer Systems, Algorithms, Heat Transfer, Data Analysis and Experimental Design

EXPERIENCE

FERGUSON BIOMECHANICS AND BIOMIMETICS LAB, Boulder, CO

2024 - 2025

Undergraduate Researcher

- Developed a workflow to enzymatically degrade articular cartilage and evaluate how the biochemical makeup at micron-scale locations relates to the localized mechanical properties of the tissue.
- Used Raman microscopy, biochemical assays, mechanical testing, and other techniques to evaluate tissue. Performed literature review and data analysis to understand how my results fit within our understanding of cartilage and osteoarthritis.

UNIVERSITY OF COLORADO BOULDER, Boulder, CO

2023 - 2024

Teaching Assistant – CAD and Fabrication

- Provided hands-on guidance to mechanical engineering students in running a lab section of a Computer Aided Design class primarily focused on building proficiency in SolidWorks
- Significantly improved my own proficiency in SolidWorks in the process of quickly identifying problems faced by my students and helping to guide them to the solutions

PROJECTS

NEXT GENERATION FLUID MANAGEMENT SYSTEM

Senior Capstone Project – Hologic Inc.

2025-2026

- With a team of 10 engineers contracted through Design Center Colorado, I designed, manufactured, and programmed a device to monitor and positively control intrauterine fluid pressure during hysteroscopic procedures
- As systems engineer, I have spearheaded the technical implementation by defining clearly defined interfaces across mechanical, electrical, and software systems, and assembled these into a single overarching system

AUTONOMOUS BEVERAGE PONG ROBOT

Mechatronics and Robotics II

2026

- On a graduate-level engineering team faced with an open-ended robotics challenge, developed a turret to autonomously launch ping-pong ball projectiles into “Solo cups.” Implementation includes mechanical hardware, computer vision, real-time aiming calculations, and UI. Managed design for manufacturing constraints with outsourced sheet metal components.
- Led computer-vision pipeline on Raspberry Pi for cup identification, target selection logic, and trajectory prediction. Contributed to mechanical design with emphasis on design for sheet metal fabrication, tolerance, and assembly requirements

AUTONOMOUS RACING ROBOT

Mechatronics and Robotics

2025

- With a team, designed and fabricated a mobile robot to compete in a “Mario Kart”-style, time trial race, including collecting and sorting icosahedral dice for time bonuses, and autonomously reacting to the behavior of the opposing robot
- Implemented motor control, sensor integration, and microcontroller-based control logic to achieve a prescribed design problem in a competition-driven setting

DRILL-POWERED BICYCLE

Component Design

2025

- With a team of 6 engineers, designed a drill-powered bicycle from concept sketches, through CAD modeling, machining, welding, and assembly, resulting in a vehicle capable of competing in a 30-minute endurance race.
- As manufacturing engineer, I was responsible for creating and tolerancing engineering drawings, fabricating all parts for drill plate, welding the steel frame, and helping to assemble the completed vehicle—then piloting it in the race

TECHNICAL SKILLS

- SolidWorks, Onshape, Bild-based version control
- C/C++, Object Oriented Programming, embedded control systems
- MATLAB, Data Analysis and Interpretation
- Microcontrollers; Raspberry Pi, Arduino
- Mechanical Testing, Nanoindentation
- Wet Lab Experience; Raman Microscopy, Biochemical Assay, Specimen Dissection, Experimental Workflow
- Rapid Prototyping: 3D Printing, Laser Cutting, Milling, Welding, Turning